Filter banks localization

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Contents

1 EnVar formulation

1.1 Standard EnVar formulation

For an ensemble N backgrounds $\left\{\mathbf{x}^1, \dots, \mathbf{x}^N\right\}$, with $\mathbf{x}^k \in \mathbb{R}^n$, the standard EnVar formulation is given by:

$$\mathbf{B} = \mathbf{X}\mathbf{X}^\mathrm{T} \circ \mathbf{L} \tag{1}$$

where:

- $\mathbf{B} \in \mathbb{R}^{n \times n}$ is the resulting background error covariance.

- $\mathbf{X} \in \mathbb{R}^{n \times N}$ contains the normalized ensemble perturbations:

$$X_{ij} = \frac{1}{\sqrt{N-1}} \left(x_i^j - \langle x_i \rangle \right) \tag{2}$$

with $\langle x_i \rangle$ denoting the ensemble mean:

$$\langle x_i \rangle = \frac{1}{N} \sum_{j=1}^{N} x_i^j \tag{3}$$

- $\mathbf{L} \in \mathbb{R}^{n \times n}$ is the localization matrix.
- $\circ$ denotes a Schur product between matrices (element-by-element).

The square-root expression of (1), transforming the control vector $\mathbf{v} \in \mathbb{R}^m$ into a control increment $\delta\mathbf{x} \in \mathbb{R}^n$, is given by:

$$\delta\mathbf{x} = \sum_{j=1}^{N} \mathbf{X}_{:j} \circ \left(\mathbf{U}\mathbf{v}_j \right) \tag{4}$$

where:

- $\mathbf{X}_{:j} \in \mathbb{R}^n$ is the j$^{\text{th}}$ column of $\mathbf{X}$
- $\mathbf{U} \in \mathbb{R}^{n \times m}$ is a square-root of the localization $\mathbf{L}$ such that $\mathbf{U}\mathbf{U}^{\mathrm{T}} = \mathbf{L}$.
- $\mathbf{v}_j \in \mathbb{R}^{m/N}$ is the j$^{\text{th}}$ subvector of $\mathbf{v}$.

1.2 Generalized EnVar formulation

The standard EnVar formulation can be generalized by introducing two linear transforms $\mathbf{F} \in \mathbb{R}^{n \times \widehat{n}}$ and $\mathbf{F}^\dagger \in \mathbb{R}^{\widehat{n} \times n}$:

$$\widehat{\mathbf{B}} = \mathbf{F} \left(\left(\mathbf{F}^\dagger \mathbf{X} \right) \left(\mathbf{F}^\dagger \mathbf{X} \right)^{\mathrm{T}} \circ \widehat{\mathbf{L}} \right) \mathbf{F}^{\mathrm{T}} \tag{5}$$

$$\delta\mathbf{x} = \mathbf{F} \sum_{j=1}^{N} \left(\mathbf{F}^\dagger \mathbf{X}_{:j} \right) \circ \left(\widehat{\mathbf{U}} \widehat{\mathbf{v}}_j \right) \tag{6}$$

where:

- $\widehat{\mathbf{L}} \in \mathbb{R}^{\widehat{n} \times \widehat{n}}$ is the localization matrix in the transformed space.
- $\widehat{\mathbf{U}} \in \mathbb{R}^{n \times \widehat{m}}$ is a square-root of the localization $\widehat{\mathbf{L}}$ such that $\widehat{\mathbf{U}}\widehat{\mathbf{U}}^{\mathrm{T}} = \widehat{\mathbf{L}}$.
- $\widehat{\mathbf{v}}_j \in \mathbb{R}^{\widehat{m}/N}$ is the j$^{\text{th}}$ subvector of $\widehat{\mathbf{v}} \in \mathbb{R}^{\widehat{m}}$.

Let's analyse the asymptotic values if the ensemble size grows to infinity:

- The localization matrices $\mathbf{L}$ and $\widehat{\mathbf{L}}$ should converge to $\mathbf{1}_n$ and $\mathbf{1}_{\widehat{n}}$, respectively, where $\mathbf{1}_n \in \mathbb{R}^{n \times n}$ is a matrix filled with 1.

- As a consequence:

$$\lim_{N \to \infty} \mathbf{B} = \mathbf{X}\mathbf{X}^{\mathrm{T}} \tag{7}$$

$$\lim_{N \to \infty} \widehat{\mathbf{B}} = \left(\mathbf{F}\mathbf{F}^{\dagger}\mathbf{X}\right)\left(\mathbf{F}\mathbf{F}^{\dagger}\mathbf{X}\right)^{\mathrm{T}} \tag{8}$$

- To ensure consistency, linear transforms $\mathbf{F}$ and $\mathbf{F}^{\dagger}$ should not have an impact on the asymptotic background error covariance:

$$\lim_{N \to \infty} \mathbf{B} = \lim_{N \to \infty} \widehat{\mathbf{B}} \tag{9}$$

  so

$$\mathbf{F}\mathbf{F}^{\dagger}\mathbf{X} = \mathbf{X} \tag{10}$$

  Thus, $\mathbf{F}^{\dagger}$ should be a right-inverse of $\mathbf{F}$ in the ensemble perturbations space.

The content of $\mathbf{F}$ and $\mathbf{F}^{\dagger}$ can be very diverse:

- A change of variables: for instance towards a better-suited humidity variable.
- A inverse balance operator as in Clayton *et al.* (2012), where the localization is applied on unbalanced variables and the balance operator is finally applied to get the increment.
- A change of resolution as in Kleist and Ide (2014), where the ensemble resolution is lower than the final increment one. In this case, $\mathbf{F}$ is an interpolation from low to high resolution, and the low-resolution perturbations $\mathbf{X}_{\mathrm{LR}}$ can be thought as the application of a coarsening operator $\mathbf{F}^{\dagger}$ on virtual high-resolution perturbations $\mathbf{X}_{\mathrm{HR}}$ representing the same field:

$$\mathbf{F}\mathbf{X}_{\mathrm{LR}} = \mathbf{F}\mathbf{F}^{\dagger}\mathbf{X}_{\mathrm{HR}} = \mathbf{X}_{\mathrm{HR}} \tag{11}$$

2 General filter banks

2.1 The band-pass framework

We define a series of P band-pass filters $\left\{\mathbf{H}_1, \dots, \mathbf{H}_P\right\}$, where $\mathbf{H}_p \in \mathbb{R}^{n \times n}$. The linear transform $\mathbf{F}^{\dagger}$ is defined as:

$$\mathbf{F}^{\dagger} = \begin{pmatrix} \mathbf{H}_1 \\ \mathbf{H}_2 \\ \vdots \\ \mathbf{H}_P \end{pmatrix} \tag{12}$$

and its left-inverse $\mathbf{F}$ as:

$$\mathbf{F} = \left(\mathbf{I}_n \mathbf{I}_n \dots \mathbf{I}_n\right) \tag{13}$$

Thus, $\mathbf{F}\mathbf{F}^\dagger$ is the sum of the band-pass filters:

$$\mathbf{F}\mathbf{F}^\dagger = \sum_{p=1}^{P} \mathbf{H}_p \tag{14}$$

and the size of the decomposition space (i.e. the number of rows of $\mathbf{F}^\dagger$) is $\widehat{n} = Pn$. For $1 \leq p \leq P-1$, the band-pass filters $\mathbf{H}_p$ are defined independently. However, to ensure that $\mathbf{F}\mathbf{F}^\dagger = \mathbf{I}_n$, the last band-pass filter $\mathbf{H}_P$ is defined as the complement of the sum of all the previous band-pass filters:

$$\mathbf{H}_P = \mathbf{I}_n - \sum_{p=1}^{P-1} \mathbf{H}_p \tag{15}$$

We also define the filtered components $\widehat{\mathbf{x}}_p \in \mathbb{R}^n$ as:

$$\widehat{\mathbf{x}}_p = \mathbf{H}_p \mathbf{x} \tag{16}$$

where $\mathbf{x} \in \mathbb{R}^n$ is the input vector.

2.2 Recursive construction

In this section, we build the band-pass filters $\mathbf{H}_p$ recursively from a series of $P-1$ filters $\left\{\mathbf{G}_1, \dots, \mathbf{G}_{P-1}\right\}$, where $\mathbf{G}_p \in \mathbb{R}^{n \times n}$:

- The initial residual is given by the input state: $\overline{\mathbf{x}}_0 = \mathbf{x}$
- At each iteration $1 \leq p \leq P-1$:
  - The filter $\mathbf{G}_p$ is applied to the previous residual $\overline{\mathbf{x}}_{p-1} \in \mathbb{R}^n$ to produce the new filtered component $\widehat{\mathbf{x}}_p \in \mathbb{R}^n$:

$$\widehat{\mathbf{x}}_p = \mathbf{G}_p \overline{\mathbf{x}}_{p-1} \tag{17}$$

  - The new residual is computed from $\widehat{\mathbf{x}}_p$:

$$\begin{aligned} \overline{\mathbf{x}}_p &= \overline{\mathbf{x}}_{p-1} - \widehat{\mathbf{x}}_p \\ &= \left(\mathbf{I}_n - \mathbf{G}_p\right) \overline{\mathbf{x}}_{p-1} \end{aligned} \tag{18}$$

This recursive construction is illustrated in Figure 1. Integrating recursively, the residual at iteration p becomes:

$$\begin{aligned} \overline{\mathbf{x}}_p &= \left(\mathbf{I}_n - \mathbf{G}_p\right)\left(\mathbf{I}_n - \mathbf{G}_{p-1}\right) \dots \left(\mathbf{I}_n - \mathbf{G}_1\right) \mathbf{x} \\ &= \prod_{q=1}^{p} \left(\mathbf{I}_n - \mathbf{G}_q\right) \mathbf{x} \end{aligned} \tag{19}$$

and from equation (17), we obtain the filtered component at iteration p:

$$\widehat{\mathbf{x}}_p = \mathbf{G}_p \prod_{q=1}^{p-1} \left(\mathbf{I}_n - \mathbf{G}_q \right) \mathbf{x} \tag{20}$$

Combining equations (16) with (20), we get an equivalence between $\mathbf{H}_p$ and $\mathbf{G}_p$ for $1 \leq p \leq P-1$:

$$\boxed{\mathbf{H}_p = \mathbf{G}_p \prod_{q=1}^{p-1} \left(\mathbf{I}_n - \mathbf{G}_q \right)} \tag{21}$$

The last filtered component $\widehat{\mathbf{x}}_P$ is actually given by the last residual $\overline{\mathbf{x}}_{P-1}$

$$\widehat{\mathbf{x}}_P = \mathbf{H}_P \mathbf{x} = \overline{\mathbf{x}}_{P-1} \tag{22}$$

So equation (19) gives another expression of equation (15):

$$\boxed{\mathbf{H}_P = \prod_{q=1}^{P-1} \left(\mathbf{I}_n - \mathbf{G}_q \right)} \tag{23}$$

2.3 Inverse formula

To inverse equation (21), a preliminary step is necessary. We start by recursively showing that for $p > 1$:

$$\mathbf{I}_n - \sum_{q=1}^{p-1} \mathbf{H}_q = \prod_{q=1}^{p-1} \left(\mathbf{I}_n - \mathbf{G}_q \right) \tag{24}$$

The initial case $p = 2$ is obvious: since $\mathbf{H}_1 = \mathbf{G}_1$ (equation (21)), then $\mathbf{I}_n - \mathbf{H}_1 = \mathbf{I}_n - \mathbf{G}_1$. We now assume that equation (24) holds for a given $p > 1$. As a consequence:

$$\begin{aligned}
\mathbf{I}_n - \sum_{q=1}^{p} \mathbf{H}_q &= \mathbf{I}_n - \sum_{q=1}^{p-1} \mathbf{H}_q - \mathbf{H}_p \\
&= \prod_{q=1}^{p-1} \left(\mathbf{I}_n - \mathbf{G}_q \right) - \mathbf{G}_p \prod_{q=1}^{p-1} \left(\mathbf{I}_n - \mathbf{G}_q \right) \\
&= \left(\mathbf{I}_n - \mathbf{G}_p \right) \prod_{q=1}^{p-1} \left(\mathbf{I}_n - \mathbf{G}_q \right) \\
&= \prod_{q=1}^{p} \left(\mathbf{I}_n - \mathbf{G}_q \right) \qquad (25)
\end{aligned}$$

so equation (24) also holds for $p+1$. We can conclude that (24) holds for any $p > 2$. Inserting it into equation (21), we get:

$$\mathbf{H}_p = \mathbf{G}_p \left(\mathbf{I}_n - \sum_{q=1}^{p-1} \mathbf{H}_q \right) \tag{26}$$

If $\left(\mathbf{I}_n - \sum_{q=1}^{p-1} \mathbf{H}_q\right)$ has a right inverse, then the filter $\mathbf{G}_p$ can be deduced from the band-pass filters:

$$\boxed{\mathbf{G}_p = \mathbf{H}_p \left(\mathbf{I}_n - \sum_{q=1}^{p-1} \mathbf{H}_q\right)^{-1}} \tag{27}$$

2.4 Embedded filters

The special case of embedded filters arises when for any $q > p$:

$$\mathbf{G}_p \mathbf{G}_q = \mathbf{G}_q \mathbf{G}_p = \mathbf{G}_q \tag{28}$$

The corresponding band-pass filters can then be expressed for $1 \leq p \leq P-1$ as:

$$\begin{aligned}
\mathbf{H}_p &= \mathbf{G}_p \prod_{q=1}^{p-1} \left(\mathbf{I}_n - \mathbf{G}_q\right) \\
&= \left(\mathbf{G}_p - \mathbf{G}_p \mathbf{G}_{p-1}\right) \prod_{q=1}^{p-2} \left(\mathbf{I}_n - \mathbf{G}_q\right) \\
&= \left(\mathbf{G}_p - \mathbf{G}_{p-1}\right) \prod_{q=1}^{p-2} \left(\mathbf{I}_n - \mathbf{G}_q\right)
\end{aligned} \tag{29}$$

and since for each factor of $\prod_{q=1}^{p-2} \left(\mathbf{I}_n - \mathbf{G}_q\right)$:

$$\begin{aligned}
\left(\mathbf{G}_p - \mathbf{G}_{p-1}\right)\left(\mathbf{I}_n - \mathbf{G}_q\right) &= \mathbf{G}_p - \mathbf{G}_{p-1} - \left(\mathbf{G}_p \mathbf{G}_q - \mathbf{G}_{p-1} \mathbf{G}_q\right) \\
&= \mathbf{G}_p - \mathbf{G}_{p-1} - \left(\mathbf{G}_q - \mathbf{G}_q\right) \\
&= \mathbf{G}_p - \mathbf{G}_{p-1}
\end{aligned} \tag{30}$$

we finally get:

$$\boxed{\mathbf{H}_p = \mathbf{G}_p - \mathbf{G}_{p-1}} \tag{31}$$

As a consequence, the construction of a sequence of embedded filters can done in parallel, as illustrated in Figure 2.

3 Real-case filters

3.1 Low-pass and high-pass filters

For a sequence of filters, we assume that there is an invertible transform $\mathbf{S} \in \mathbb{R}^{n \times n}$ such that every component can be expressed as:

$$\mathbf{G}_p = \mathbf{S}^{-1} \mathbf{D}_p \mathbf{S} \tag{32}$$

where $\mathbf{D}_p \in \mathbb{R}^{n\times n}$ is a diagonal matrix with diagonal elements between 0 and 1. The spectrum $\mathbf{d}_p \in \mathbb{R}^n$ contains all these diagonal elements: $D_{p,ii} = d_{p,i}$.

For a sequence of low-pass filters, all the spectra start from 1 and are monotonically decreasing. Symmetrically for a sequence of high-pass filters, all the spectra are monotonically increasing and end up at 1. In both cases, for any $p > 1$ and any element: $d_{p,i} \geq d_{p-1,i}$, as illustrated in Figure 3.

3.2 High-pass filters built from low-pass filters

Often in practice, only low-pass filters $\mathbf{G}^l_p$ are actually implemented, through a diffusion process or an equivalent smoother. However, the high-pass filter $\mathbf{G}^h_p$ can be built as the complement of the low-pass filter sorted in opposite order $\mathbf{G}^l_{P-p}$:

$$\mathbf{G}^h_p = \mathbf{I}_n - \mathbf{G}^l_{P-p} \tag{33}$$

Updating equation (21), the band-pass filter $\mathbf{H}^h_p$ corresponding to the high-pass filters is given for $1 \leq p \leq P-1$ by:

$$\begin{aligned}\mathbf{H}^h_p &= \mathbf{G}^h_p \prod_{q=1}^{p-1} \left(\mathbf{I}_n - \mathbf{G}^h_q\right) \\ &= \left(\mathbf{I}_n - \mathbf{G}^l_{P-p}\right) \prod_{q=1}^{p-1} \mathbf{G}^l_{P-q}\end{aligned} \tag{34}$$

Figure 4 shows explicit low-pass filters and their complementary high-pass filters counterparts. The corresponding band-pass filters are also displayed in both cases. We can see that the band-pass filters have roughly the same structure in both cases, but are not identical.

3.3 Embedded filters

If a sequence of low-pass or high-pass filters is embedded, equation (28) can be simplified for $q < p$ into:

$$d_{p,i} d_{q,i} = d_{q,i} \tag{35}$$

which is only true if:

$$\begin{cases} \text{either} & d_{q,i} = 0 \\ \text{or} & d_{p,i} = 1 \end{cases} \tag{36}$$

A sequence of embedded low-pass filters is illustrated on Figure 5. If the sequence of low-pass filters is embedded, it is easy to see that the sequence of complementary high-pass filters is embedded too, and the corresponding band-pass filters can be derived from equation (34):

$$\begin{aligned}\mathbf{H}^h_p &= \left(\mathbf{I}_n - \mathbf{G}^l_{P-p}\right) \prod_{q=1}^{p-1} \mathbf{G}^l_{P-q} \\ &= \mathbf{G}^l_{P-p+1} - \mathbf{G}^l_{P-p}\end{aligned} \tag{37}$$

So the corresponding band-pass filters are equivalent for the low-pass filters and their high-pass complementary filters (sorted in opposite order):

$$\boxed{\mathbf{H}^h_p = \mathbf{H}^l_{P-p+1}} \tag{38}$$

It seems that the difference between a sequence of low-pass filters and the sequence of the complementary high-pass filters becomes greater when more than two bands of the corresponding band-pass filters overlap. In the case of embedded filters where only two bands overlap, this difference vanishes.

3.4 Pseudo-embedded filters

Embedded filter are appealing because they can be applied in parallel (see Figure 2). However, low-pass filters based on a diffusion process or an equivalent smoother are not able to produce this kind of filters. The corresponding convolution functions in grid-point space induced by these methods are generally positive, whereas embedded filters require negative lobes, as illustrated in Figure 6.

A possible option is to use the parallel scheme of Figure 2 with non-embedded filters. If the different components of the sequence of low-pass filters are sufficiently separated, the difference with the recursive scheme of Figure 1 is minor, as illustrated in Figure 7.

3.5 Reduced resolution filters

For a sequence of low-pass filters, it is possible to reduce the memory and computational cost by storing their output at a lower resolution. For this, we redefine the band-pass filter output space: $\mathbf{H}_p \in \mathbb{R}^{\widehat{n}_p \times n}$ where:

- $\widehat{n}_q \leq \widehat{n}_p$ for $q < p$,
- $\widehat{n}_P = n$ for the last component.

The linear transform $\mathbf{F}^\dagger$ is still defined as:

$$\mathbf{F}^\dagger = \begin{pmatrix} \mathbf{H}_1 \\ \mathbf{H}_2 \\ \vdots \\ \mathbf{H}_P \end{pmatrix} \tag{39}$$

but its left-inverse is redefined as:

$$\mathbf{F} = \left(\mathbf{S}_{P\leftarrow 1} \mathbf{S}_{P\leftarrow 2} \dots \mathbf{I}_n\right) \tag{40}$$

where $\mathbf{S}_{p\leftarrow q} \in \mathbb{R}^{\widehat{n}_p \times \widehat{n}_q}$ is an interpolator from the resolution $\widehat{n}_q$ to the resolution $\widehat{n}_p$. Thus, $\mathbf{F}\mathbf{F}^\dagger$ is the sum of the band-pass filters interpolated at full resolution:

$$\mathbf{F}\mathbf{F}^\dagger = \sum_{p=1}^{P} \mathbf{S}_{P\leftarrow p} \mathbf{H}_p \tag{41}$$

and the size of the decomposition space (i.e. the number of rows of $\mathbf{F}^\dagger$) is the sum of all the sizes of the components:

$$\widehat{n} = \sum_{p=1}^{P} \widehat{n}_p \tag{42}$$

From the filters $\mathbf{G}_p \in \mathbb{R}^{\widehat{n}_p \times n}$, the band-pass filters $\mathbf{H}_p$ are computed for $1 \leq p \leq P-1$ as:

$$\boxed{\mathbf{H}_p = \mathbf{G}_p \prod_{q=1}^{p-1} \left(\mathbf{I}_n - \mathbf{S}_{P \leftarrow q} \mathbf{G}_q\right)} \tag{43}$$

and the last component as:

$$\boxed{\mathbf{H}_P = \prod_{q=1}^{P-1} \left(\mathbf{I}_n - \mathbf{S}_{P \leftarrow q} \mathbf{G}_q\right)} \tag{44}$$

The updated recursive scheme is given by Figure 8.

For the embedded filters, the assumption (28) should include the interpolators:

$$\mathbf{G}_p \mathbf{S}_{P \leftarrow q} \mathbf{G}_q = \mathbf{S}_{p \leftarrow q} \mathbf{G}_q \mathbf{S}_{P \leftarrow p} \mathbf{G}_p = \mathbf{S}_{p \leftarrow q} \mathbf{G}_q \tag{45}$$

and the interopolators should be transitive, i.e. for $q < p$:

$$\mathbf{S}_{P \leftarrow p} \mathbf{S}_{p \leftarrow q} = \mathbf{S}_{P \leftarrow q} \tag{46}$$

If these conditions are fulfilled, the updated parallel scheme is shown in Figure 9.

References

Clayton AM, Lorenc AC, Barker DM. 2012. Operational implementation of a hybrid ensemble/4D-Var global data assimilation system at the Met Office. *Quarterly Journal of the Royal Meteorological Society* **139**: 1445–1461.

Kleist DT, Ide K. 2014. An OSSE-Based evaluation of hybrid variational-ensemble data assimilation for the NCEP GFS. Part I: System description and 3D-hybrid results. *Monthly Weather Review* **143**(2): 433–451.

Figures

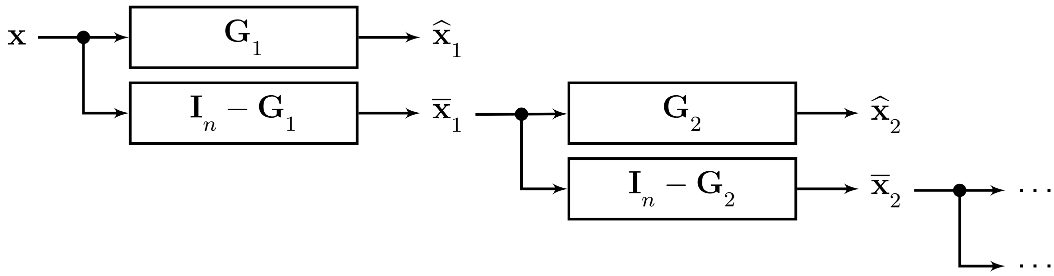

Figure 1: Recursive filter banks.

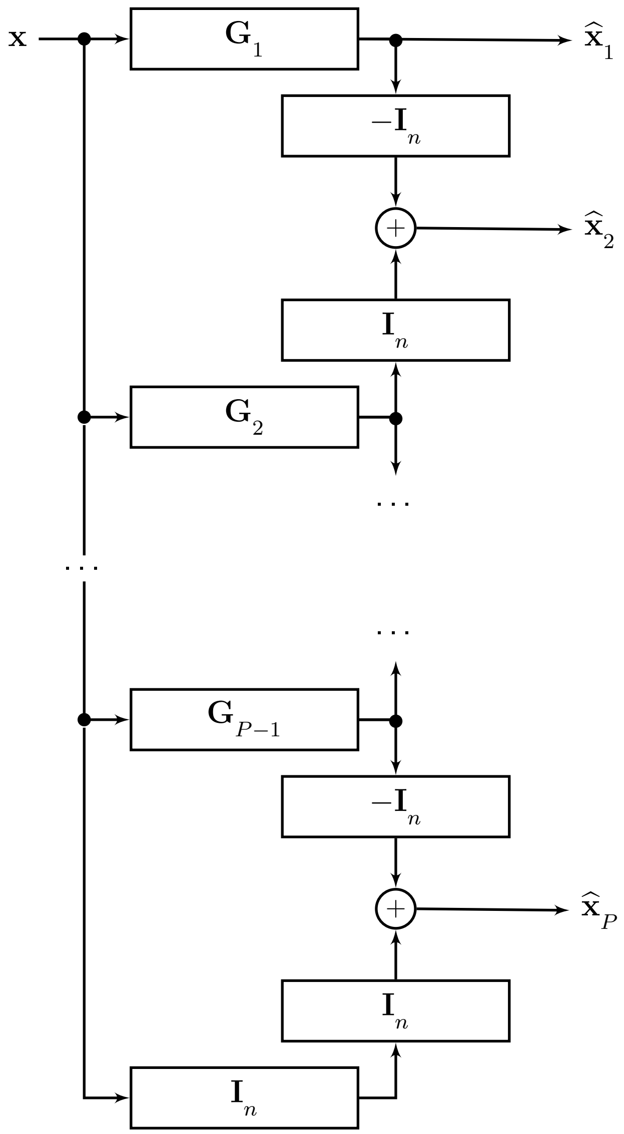

Figure 2: Parallel filter banks.

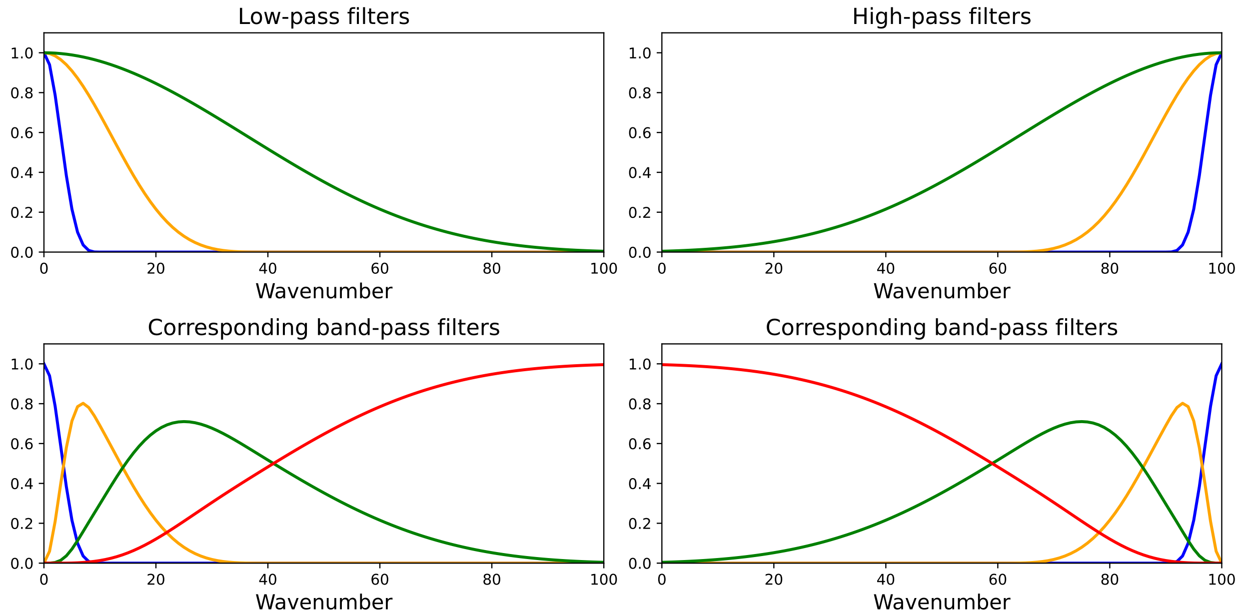

Figure 3: Spectra of low-pass filters (top left) and high-pass filters (top right), and their corresponding band-pass filters (bottom).

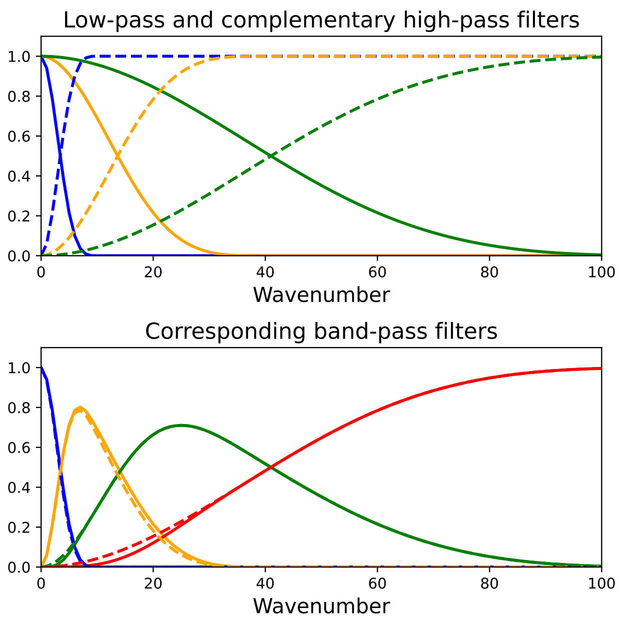

Figure 4: Spectra of low-pass filters (top, solid) and the complementary high-pass filters (top, dashed) and the corresponding band-pass filter (bottom).

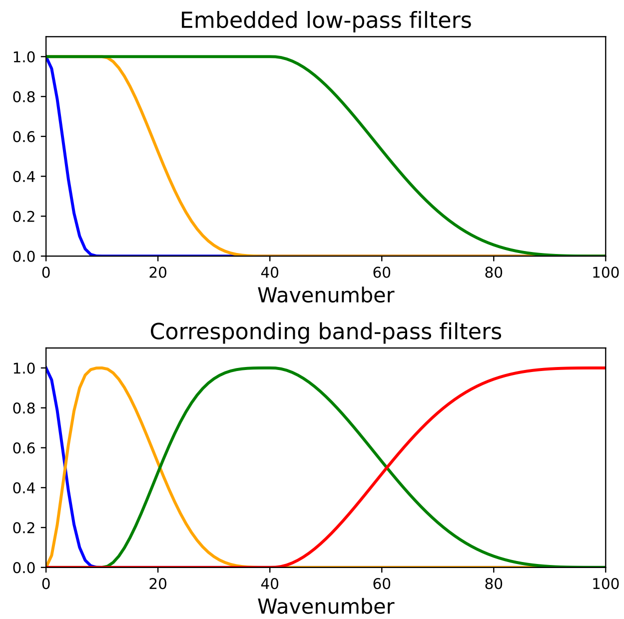

Figure 5: Spectra of embedded low-pass filters (top) and the corresponding band-pass filters (bottom).

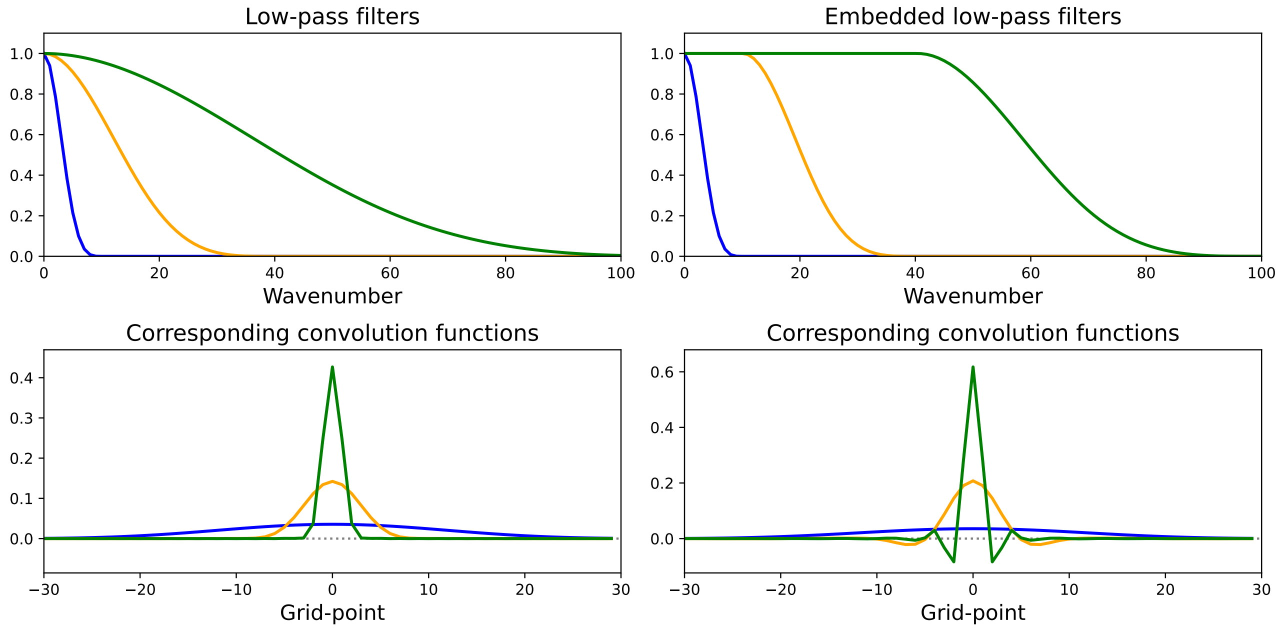

Figure 6: Spectra of low-pass filters (top left) and embedded low-pass filters (top right), and the corresponding convolution functions (bottom).

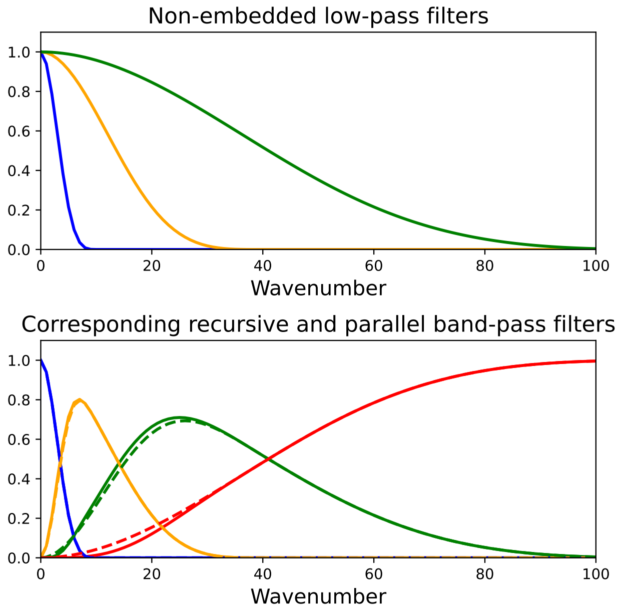

Figure 7: Spectra of low-pass filters (top) and the corresponding band-pass filters computed in recursive (bottom, solid) or parallel mode (bottom, dashed).

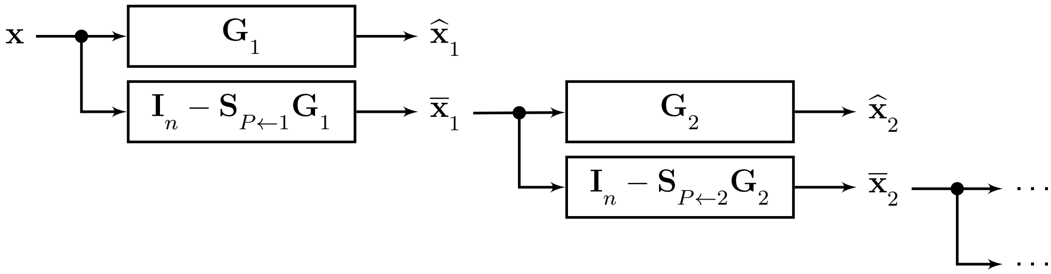

Figure 8: Recursive filter banks with reduced resolutions.

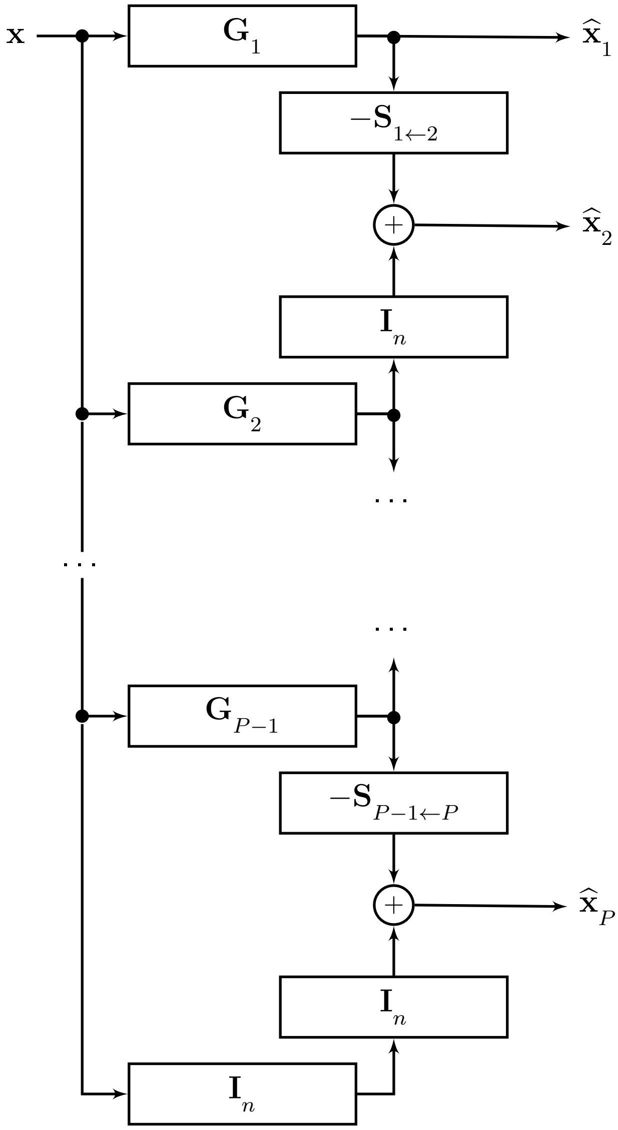

Figure 9: Parallel filter banks with reduced resolutions.